

Ex. No. : 07

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Telephony services

Aim

Implement an application to get Telephony services.

Procedure:

Step 1 : File -> NewProject

Provide the application name and Click “Next”

Step 2 : Select the target android devices

Select the minimum SDK to run the application. Click “Next”.

Step 3 : Choose the activity for the application (By default choose “Blank Activity”).

Click “Next”.

Step 4 : Enter activity name and click.

Step 5 : Edit the program.

Step 6 : Run the application, 2-ways to run the application.

1. Running through emulator

2. Running through mobile device

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <uses-permission android:name="android.permission.READ_PHONE_STATE"/>

    <application
        android:allowBackup="true"
        android:label="Telephony App"
        android:supportRtl="true"
        android:theme="@style/Theme.TelephonyServices">
        <activity android:name="com.example.telephonyservices.MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN"/>
                <category android:name="android.intent.category.LAUNCHER"/>
            </intent-filter>
        </activity>
    </application>

</manifest>
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp">

    <TextView
        android:id="@+id/textView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Fetching Telephony Info..."
        android:textSize="18sp"
        android:textColor="#000000"/>
</RelativeLayout>
```

MainActivity.kt

```
package com.example.telephonyservices

import android.Manifest
import android.content.pm.PackageManager
import android.os.Bundle
import android.telephony.TelephonyManager
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity
import androidx.core.app.ActivityCompat
import androidx.core.content.ContextCompat

class MainActivity : AppCompatActivity() {

    private val PERMISSION_REQUEST_CODE = 101

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        if (ContextCompat.checkSelfPermission(this, Manifest.permission.READ_PHONE_STATE)
            != PackageManager.PERMISSION_GRANTED
        ) {
            ActivityCompat.requestPermissions(
                this,
                arrayOf(Manifest.permission.READ_PHONE_STATE),
                PERMISSION_REQUEST_CODE
            )
        } else {
            displayTelephonyInfo()
        }
    }

    override fun onRequestPermissionsResult(
        requestCode: Int, permissions: Array<out String>, grantResults: IntArray
    ) {
        super.onRequestPermissionsResult(requestCode, permissions, grantResults)

        if (requestCode == PERMISSION_REQUEST_CODE && grantResults.isNotEmpty()
            && grantResults[0] == PackageManager.PERMISSION_GRANTED
        ) {
            displayTelephonyInfo()
        } else {
```

```

        findViewById<TextView>(R.id.textView).text = "Permission Denied"
    }
}

private fun displayTelephonyInfo() {
    val telephonyManager = getSystemService(TELEPHONY_SERVICE) as TelephonyManager

    val info = StringBuilder()
    info.append("Network Operator: ${telephonyManager.networkOperatorName}\n")
    info.append("SIM Country: ${telephonyManager.simCountryIso.uppercase()}\n")
    info.append("Phone Type: ${getPhoneTypeString(telephonyManager.phoneType)}\n")

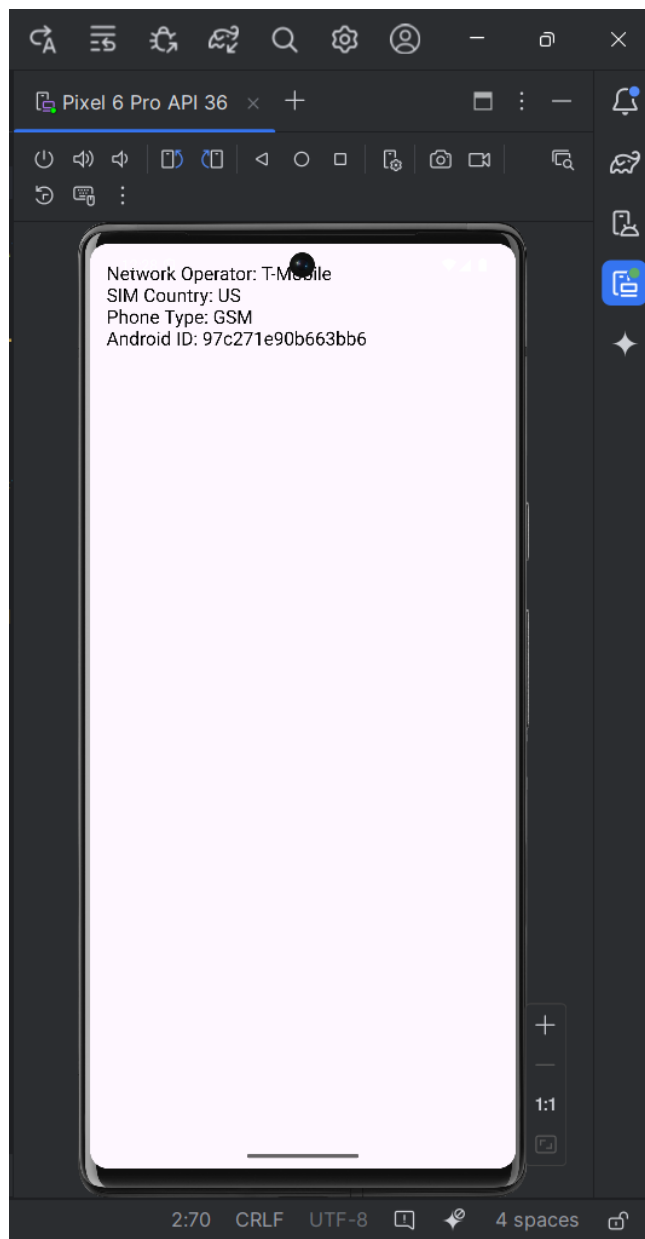
    // Instead of IMEI, use ANDROID_ID which works on all versions
    val androidId = android.provider.Settings.Secure.getString(
        contentResolver,
        android.provider.Settings.Secure.ANDROID_ID
    )
    info.append("Android ID: $androidId\n")

    findViewById<TextView>(R.id.textView).text = info.toString()
}

private fun getPhoneTypeString(type: Int): String {
    return when (type) {
        TelephonyManager.PHONE_TYPE_GSM -> "GSM"
        TelephonyManager.PHONE_TYPE_CDMA -> "CDMA"
        TelephonyManager.PHONE_TYPE_SIP -> "SIP"
        TelephonyManager.PHONE_TYPE_NONE -> "None"
        else -> "Unknown"
    }
}
}

```

Output



Result:

The Telephony Services experiment has been successfully completed